

Prevention of Timing Attack in Software Defined Named Data Network with VANETs

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Abstract—Software Defined Network (SDN) is getting popularity both from academic and industry. Lot of researches have been made to combine SDN with future Internet paradigms to manage and control networks efficiently. SDN provides better management and control in a network through decoupling of data and control plane. Named Data Networking (NDN) is a future Internet technique with aim to replace IPv4 addressing problems. In NDN, communication between different nodes done on the basis of content names rather than IP addresses. Vehicular Ad-hoc Network (VANET) is a subtype of MANET which is also considered as a hot area for future applications. Different vehicles communicate with each other to form a network known as VANET. Communication between VANET can be done in two ways (i) Vehicle to Vehicle (V2V) (ii) Vehicle to Infrastructure (V2I). Combination of SDN and NDN techniques in future Internet can solve lot of problems which were hard to answer by considering a single technique. Security in VANET is always challenging due to unstable topology of VANET. In this paper, we merge future Internet techniques and propose a new scheme to answer timing attack problem in VANETs named as Timing Attack Prevention (TAP) protocol. Proposed scheme is evaluated through simulations which shows the superiority of proposed protocol regarding detection and mitigation of attacker vehicles as compared to normal timing attack scenario in NDN based VANET.

Index Terms—Content, Interest, Software Defined Networks, Named Data Networking, Ad Hoc Networks, Vehicular Ad Hoc Networks, Timing Attack

I. INTRODUCTION

As Internet is flourishing, network communication is becoming complex and error prone. Users need better and efficient ways to communicate with each other. Due to increase in Internet applications like online games, VoIP, and video streaming, communication networks need better mechanism to manage and control the complexity of the system. Complex network system is composed of large number of switches, routers or hubs. It is the duty of the network administrator to compose and implement different protocols on large variety of networks. The main problem occurs when administrator has to implement these protocols through low stage commands which are hard and complex to remember as well as they have restricted tools to perform such actions on a network. Therefore, management and control of a network is complex and error sensitive.

Software Defined Network (SDN) [1] is a new emerging field in computer networks with aim to manage and control the network systems more easily and efficiently. SDN main function is to separate the network devices (i.e. switches, routers) from the control decisions. Its benefit is that it strikingly change the management of a network from complex to simple. There are two basic components of SDN (i) Control plane (ii) Data plan. Control plane consist of SDN controller which manages and control whole network while data plane consists of switches or routers which forward the data to other nodes. SDN controller can be programmable through simpler interface.

Named Data Networking (NDN) [2] is a new technique in computer networks with aim to overcome all the issues of IPv4 addressing. In NDN, communication between different nodes can be achieved through content names rather than their IPv4 addresses. In this process, one or more nodes become consumer or producer nodes. A consumer node is a node which needs a specific data and producer node is a node which contains that specific data and respond back to the consumer node by sending requested data. This mechanism is used to overcome the addressing limitation of IPv4 [3].

Another paradigm that is getting popular now a days is Vehicular Ad-hoc Networks (VANETs). VANET is basically a sub type of Mobile Ad-hoc Networks (MANETs)[4]. In VANET, communication is done between vehicles in two forms . (i) Vehicle to Vehicle (V2V) (ii) Vehicle to Infrastructure (V2I) [5]. In V2V, communication is done between two or more different vehicles on highway. Packets are exchanged between vehicles until the receiving vehicle received the packets. In V2I, communication is done with vehicles and road side units (RSUs). Road side units are static infrastructures which can be in form of road side signals. Packets are exchanged between vehicles and RSU until the producer vehicle received these packets. The packets exchanged in VANETs are categorized into two types (i) Safety critical packets (ii) Infotainment packets.

With the combinations of all these techniques different security attacks can mitigate easily. Timing attack is one of the attack that can be detected and mitigated using SDN and NDN in VANETs. In timing attack an attacker node exist in a network which add delay in safety critical packets.

These packets can contain information about an accident, vip protocol or traffic jam scenario.

Therefore, in this paper, by taking the advantages of SDN and NDN communication paradigm in VANETs, we proposed a mechanism to overcome the timing attack problem named as Timing Attack Prevention (TAP) protocol. In TAP, a legitimate vehicle detects an attacker vehicle while network controller is used to mitigate that detected attacker.

The rest of this paper is organized as follows. In Section II, we discussed detailed studies in terms of timing attack. Section III defines the architecture of software defined named data network with VANET. Then in section IV, we describe the main research problem which we tackled in our scheme. In section V, we presented our proposed scheme regarding timing attack problem. Section VI provides the simulation results and analysis, and finally the conclusions are presented in Section VII.

II. RELATED WORK

In this section, we discuss some of the studies which are done on software defined named data network with VANET and on timing attack problem.

Ahmad et al [6]. proposed first ever architecture of software defined vehicular network with respect of named data networking. They discussed possible combination of NDN with software defined vehicle network (SDVN). Main working of their research includes the pull based and push based forwarding mechanism in SDVN with NDN. This is only paper till now in which all three SDN, NDN and VANET techniques are combined together.

Irshad et al [7]. discussed some issues related to timing attack problem in detail. According to authors, V2V have two levels of timing attack (i) Basic level attack (ii) Extended level attack. In basic level, peer to peer (P2P) network communication suffers from this attack while in extended level, more than one attacker attacks the network. In V2I, attacker attack on authentication and for taking services.

Paolo et al [8]. proposed a scheme named VIPER: a Vehicle-to-Infrastructure Communication Privacy Enforcement Protocol. This protocol shows strong resilient to three VANET attacks: timing attack, message volume and message coding attack. Their results show that VIPER performed well in terms of message delivery time, message path length and queue occupancy.

In addition to these schemes, there exist multiple research studies on timing attack in VANET. However, all above mentioned schemes are limited to one or two future Internet paradigm and there exist no such scenario in which SDN, NDN and VANET technologies are implemented together. In this paper, we are first to our knowledge that combined SDN, NDN and VANET paradigm together and implemented a novel scheme for timing attack which not only detect but also mitigate attacker from network with the help of SDN.

III. SOFTWARE DEFINED NAMED DATA NETWORK WITH VANET

In software defined named data network with VANET architecture, a centralized controller is present which is used to manage and control whole network. Communication between different vehicles and controller is done through NDN messages. Vehicles which are interested in communication are called consumer vehicles. These vehicles broadcast the NDN request packets which are known as interest packets. In reply of consumer nodes, there exist multiple producer nodes which contain requested data in their cache. Data packets are generated same like interest packets from producers which are then sent back to consumers. In software defined named data network with VANET, each vehicle communicate with other vehicles and road side units (RSUs). These RSUs then broadcast corresponding packets to other RSUs and controller through Internet. Response of these corresponding packets follow exact same path from controller to vehicles. Also each vehicle in network have its own cache to store data [6]. As shown in Fig 1., red consumer vehicle generates interest packets which are broadcast in vehicle range. Two other vehicles and RSU2 receive these packets. On arrival of these packets each vehicle forward data packet to consumer vehicle if they have data in their cache. In case of RSU2, it forwards the packet to controller using Internet and then a response from controller is forwarded to corresponding vehicle by RSU2 using same path.

A. Traditional NDN implementation

In native NDN, when a new packet arrived it only takes one of two possible processing paths which is determined using packet type. This packet type is decoded to check whether incoming packet is interest or data packet. There exist three data structures in NDN named as pending interest table (PIT), content store (CS) and forwarding information base (FIB) [3]. If decoded packet turns out to be a data packet then first of all PIT entry of corresponding node is checked. If PIT entry matched then node caches the data in its CS and forward data to next nodes. If PIT entry does not matches then it simply discard the packet. In case of interest packet, PIT entry of corresponding node is matched against interest packet. If PIT is found then it means that this specific interest is already present in node so it simply discard the packet. If PIT is not found, it first check its CS and if no entry found in CS it forwards the packet to FIB which is responsible of forwarding it to next face. However, if CS entry found then node simply forwards the data in response of interest packet. Fig 2. shows packets processing in traditional NDN. Further studies regarding NDN implementation over VANET can be found in [9],[10],[11].

IV. TIMING ATTACK PROBLEM

As the Internet is expanding the network security is becoming more complex. Lot of researches have been done both on traditional wired and wireless networks in terms of security of networks. Through these researches lot of

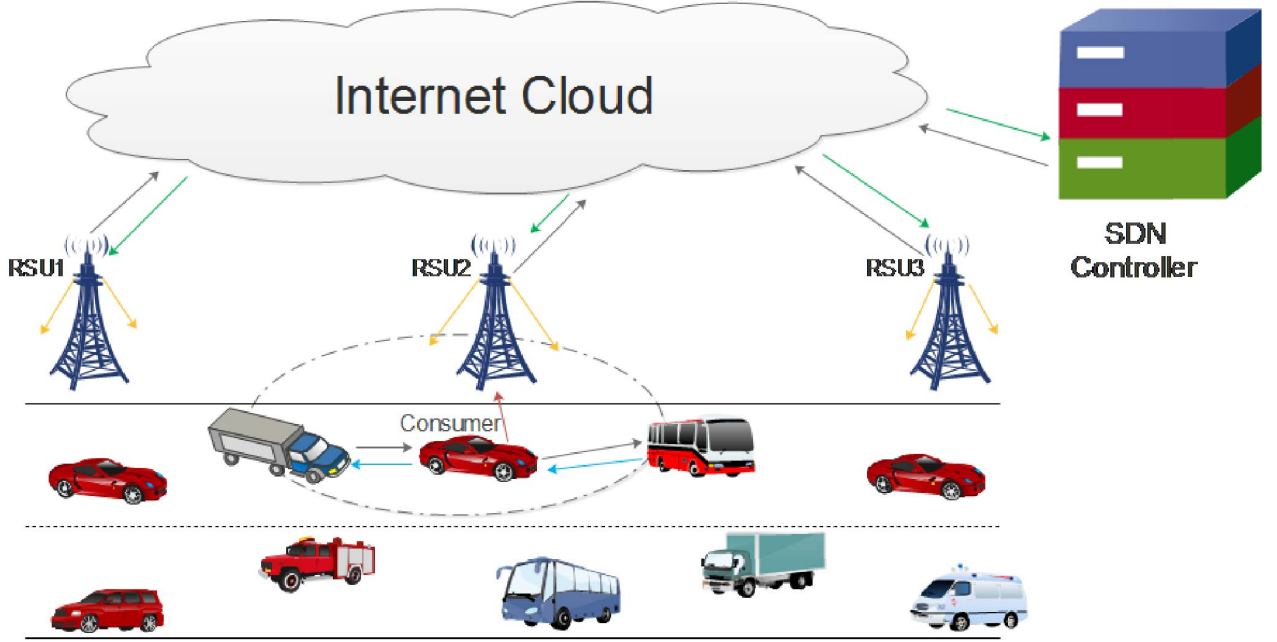


Fig. 1. Software defined named data network with VANET architecture

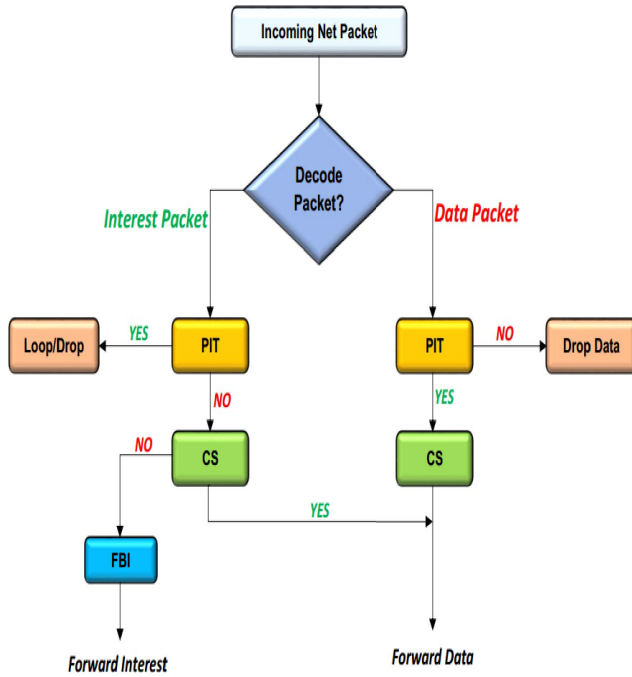


Fig. 2. Packet processing in NDN

attacks are mitigated but some of them are still pending in wireless due to its unstable topologies. In VANETs, there are two type of applications (i) Safety critical applications (ii) Infotainment applications [4]. Safety critical applications are those application which require data in real time. They contain

sensitive data which must be received to vehicles without any delay. These applications contain data about the situation of highway like accident information, VIP protocol movement and traffic jam. Transmitting and receiving data in VANET are important to achieve data integrity and security. Timing attack which is relatively a new attack in VANETs contains one or more attacker vehicles. These attacker vehicles do not forward the safety critical data to other vehicles on right time but they add some extra delays in packets in terms of time-slots [7]. However, other vehicles which are near to attacker vehicle received these data packets after they actually require it. Fig 3 graphically illustrates timing attack scenario. In this scenario, an accident is occurred between vehicle A and B. Vehicle B broadcast this critical information in terms of emergency packet in a network. Therefore, when this emergency packet is received on malicious yellow vehicle it does not transmit it to the vehicle C at the right time. But it add some extra delay in terms of time-slots in the packet so whenever vehicle C receives these emergency packet it is on the spot C_1 where the accident has occurred [12],[13].

V. PROPOSED SCHEME

In this section, we proposed a scheme for timing attack problem named as Timing Attack Prevention (TAP) protocol to detect and mitigate attacker vehicle in software defined named data network with VANET.

A. Timing Attack Prevention (TAP)

Detection of the attacker vehicles and than mitigating them using the functionality of controller in a network is the responsibility of TAP scheme. Algorithm 1 shows the processing of

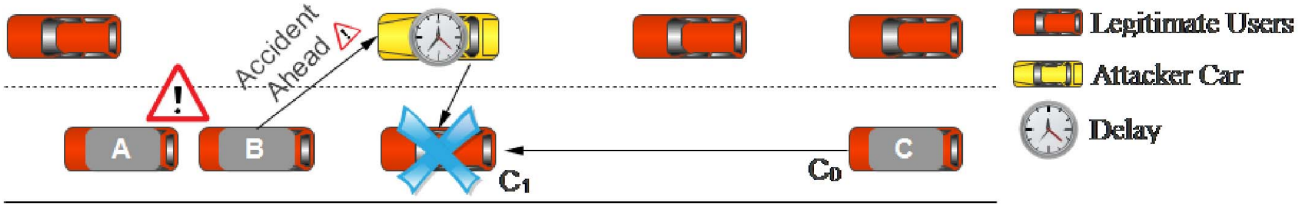


Fig. 3. Timing attack scenario

Algorithm 1: Send/Receive Emergency Packet

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1: OnSendEmergencyPacket[vID, getCoor, PAT, PPAT, PGT]
2: vID ← getNodeID()
3: xCoordinates ← getX()
4: yCoordinates ← getY()
5: setCoor(xCoordinates, yCoordinates)
6: PAT ← getPacketArrivalTime()
7: PPAT ← getPerviousPacketArrivalTime()
8: PGT ← getPacketGenerationTime()

9: OnReceivedEmergencyPacket [vID, getCoor, PAT, PPAT, PGT]
10: If (vID != Controller_Defaultler_List) then
11:   D ← calculateDistance(receiver(getCoor), sender(getCoor))
12:    $\theta \leftarrow \frac{D}{S}$ 
13:   v ← PAT -  $\theta$ 
14:   if (v != PPAT) then
15:     Forward vID to Controller
16:   else
17:     Forward Emergency Packet to Face
18:   end if
19: else
20:   Discard Packet
21: end if

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send and receive emergency packet for safety critical applications. In order to send emergency packets to other vehicles, first coordinates and vehicle ID of corresponding vehicle are set in the packet fields i.e x coordinates, y coordinates and vID fields respectively (Lines 1-5). After that vehicle gets packet arrival time (PAT) and previous packet arrival time (PPAT). But if corresponding vehicle is consumer and sending the packet for very first time in the network than PAT, PPAT values will be 0 (Lines 6-7). Now in the last packet generation time (PGT) is get using clock of the vehicle (Line 8). The main contribution of the proposed scheme (TAP) is to mitigate the attacker from the network. However, when a new vehicle received emergency packet it first checks the Controller-Defaulter-List. This list will be populated using control messages. Packet received by a vehicle contains the previous sender vehicle id. Now this vehicle id is matched against the Controller-Defaulter-List which was already stored in cache (Line 10). If results matched than automatically vehicle dropped the latest packet and perform no further action (Lines 19-20). It shows that this previously received packet is from such a vehicle which was already detected as a attacker by controller. If no results are matched then this means that previously received

packet is from such a vehicle which is not attacker or not detected as a attacker. So in this case after matching with attacker list, the detection process start. In detection process, a vehicle first calculates distance using its x,y and sender's x,y coordinates (Line 11). In next step, after distance calculation vehicle finds out the time period in which a packet is arrived from a source to destination using signal propagation speed which is equal to 3.0×10^8 m/s. is a time period which can be get by dividing distance (D) over signal speed (S) (Line 12). After that this time period value is minus from packet arrival time (PAT) to get previous vehicle arrival time to check whether previous vehicle added any delay in packet or not (Line 13-14). If the value of previous vehicle arrival time does not matches with the value of (PPAT) in packet then it means that previous vehicle is attacker and added delay in the packet. This packet is now useless as it comes to vehicles after they actually require it. Therefore, previous vehicle id is then forwarded to controller so that it stores attacker id in its defaulter list (Line 15) and if no delay is added then vehicle simply forward the emergency packet to other vehicles as this means that previous vehicle is not an attacker vehicle (Lines 16,17).

VI. PERFORMANCE EVALUATIONS

We use ndnSim [14] using NS3 and SUMO [15] to simulate proposed scheme and considered protocol in vehicular ad-hoc network scenario. We randomly generate mobility of 64 vehicles in a network. These vehicles move in a constant speed. Therefore no vehicle can overtake others. Some simulation parameters regarding TAP is shown in Table 1.

In our scenario, we assume that 70% of vehicles are legitimate users while remaining 30% are attacker vehicles. To investigate average delay and duplicate emergency packets we vary the number of attackers in a network. Attackers proportion are set as 5%,10%,15%,20%,25%. We simulate our simulations for 300s. We also use IEEE 802.11a, since IEEE 802.11p is not supported by ndnSIM v2.0 [16]. In our simulations, we evaluated percentage of attackers that generated delays, control broadcast ratio and duplicate emergency packets. To evaluate the performance of both proposed and normal VNDN with attacker we have considered following metrics:

Average delay: Average delay can be measured by adding all delays. A delay can be define as amount of time taken by

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Number of vehicles	64
Type of vehicles	Mobile
Area (m * m)	1000 * 1000
Mobility	Uniform
Simulation time	300s
No. of attackers	5%,10%,15%,20%,25%
Emergency packet timeout	20s
No. of legitimate vehicles	70%
Mobility model used	Random Direction
MAC Layer	IEEE 802.11a
Packet size	400 bit
Tx power	0.0091 mW
Traffic Generator Tool	SUMO

packet to travel across network from source to destination.

Control packets broadcast ratio: This ratio is equal to total number of control packets generated by controller whenever an attacker vehicle is identified.

Duplicate Emergency packets (DEP): Duplicate emergency packets are those packets which attacker vehicles generate after adding time slots.

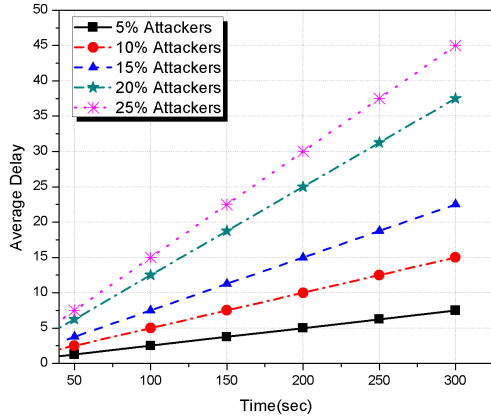


Fig. 4. Average delay of attacker vehicles

A. Simulation Results

Fig 4. represents average delay according to the attacker vehicles in a normal VNDN. As simulation time increases in normal VNDN with attacker implementation, attacker vehicles generate more delay. This is mainly due to the fact that there exist no such mechanism to prevent attackers from adding time-slots in packets.

Fig 5. shows the ratio of control packets. These packets are exchanged in a network when an attacker is identified by a legitimate vehicle. Control packets broadcast ratio is directly proportional to number of attackers. These control packets include attacker vehicle ids that were identified in detection phase. TAP mitigation phase done on each vehicles whenever a new packet is received.

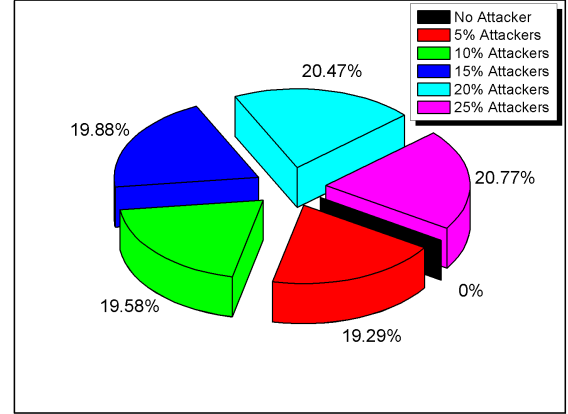


Fig. 5. Control packets broadcast ratio according to attackers

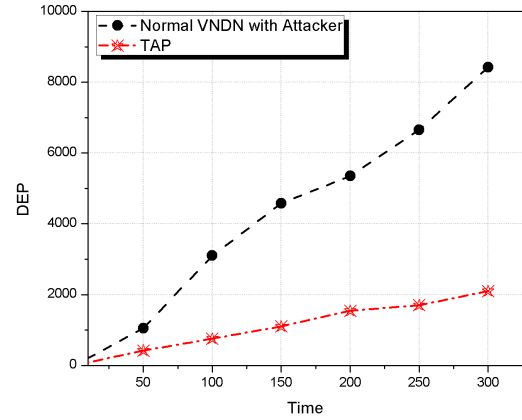


Fig. 6. Duplicate emergency packets with respect to time

Fig 6. shows number of duplicate emergency packets. These packets are generated whenever attacker vehicles add delays in packets. In normal VNDN with attackers, large number of duplicate emergency packets were broadcast in network. While in case of TAP, this number decreased significantly due to controller default list. This list is checked by every vehicle before starting detection phase. These vehicles match the coming packet vehicle id field to controller default list and performs action according to the function of receive emergency packet in algorithm 1.

VII. CONCLUSION

In this paper, we implemented software defined named data networking with VANET and proposed technique to tackle timing attack in safety critical applications. In TAP, we first detect a vehicle to check whether corresponding vehicle is an attacker or not. When an attacker vehicle is detected,

SDN controller is used to mitigate it with the help of controller default list. This mechanism ensured that no delayed emergency packets forwarded in the network. This detection process is always done on next hop node from attacker vehicle. Our simulation results show that increased delay is due to increased attacker vehicles in a network and also control packets broadcast ratio increases with increase of attacker vehicles. Duplicate emergency messages are also decreased in a network with the help of TAP protocol while in case of normal VNDN timing attack they increases significantly.

As a future work, we will work on other security issues of VANET with support of NDN and SDN controller.

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